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Secure Distribution Venture Pvt. Ltd.

What Is Renewable Energy Funding ?

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Executive Institutional Summary & Masterclass Overview

Renewable Energy Funding represents the specialized project finance, structured credit, and capital market mechanisms designed to fund the development, construction, acquisition, and refinancing of clean energy generation assets—including solar photovoltaic (PV) plants, onshore and offshore wind farms, hybrid solar-wind systems, Battery Energy Storage Systems (BESS), green hydrogen facilities, and biomass/small-hydro installations across India. Characterized by high upfront capital intensity, near-zero marginal variable cost, and long-term predictable tariff cash flows governed by 25-year Power Purchase Agreements (PPAs), renewable energy assets are fundamentally underwritten using non-recourse or limited-recourse Special Purpose Vehicle (SPV) project finance structures.

This comprehensive white paper manual serves as an authoritative textbook for finance professionals, corporate treasurers, investment bankers, risk managers, and credit analysts. By exploring statutory frameworks under the Reserve Bank of India (RBI) Project Finance Guidelines, Central Electricity Regulatory Commission (CERC) regulations, Electricity (Promoting Renewable Energy Through Green Energy Open Access) Rules, and the Electricity Act 2003, this document provides complete mastery over project debt sizing, capacity utilization factor (CUF) P50/P90 generation modeling, Trust and Retention Account (TRA) cash flow waterfalls, Section 32 accelerated depreciation tax shields, and risk mitigation architecture.

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Module 1: Foundations of Renewable Energy Funding in India

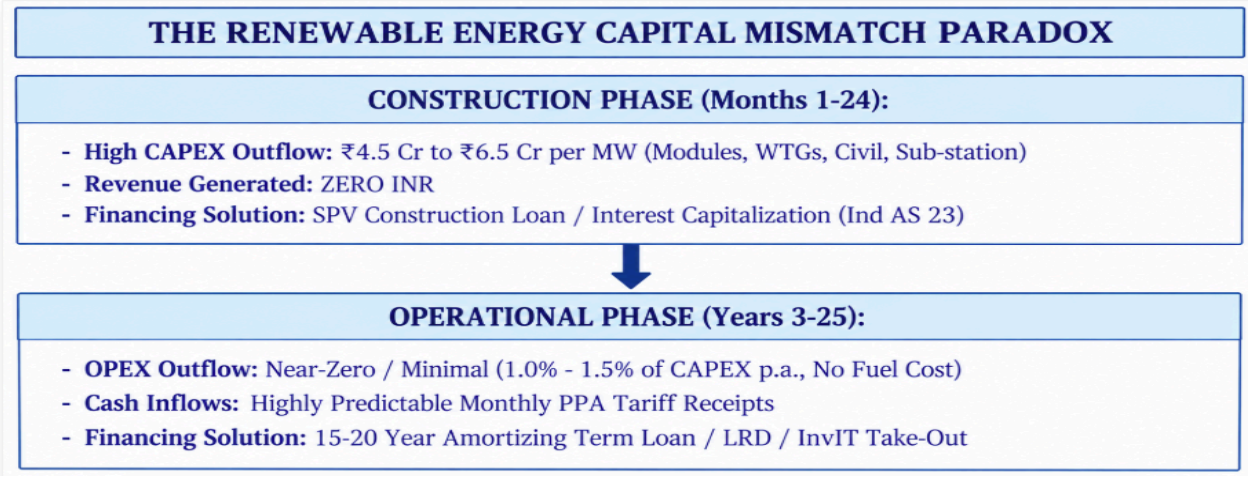
1.1 The Core Paradigm: SPV Non-Recourse Debt vs. Corporate Balance Sheet Financing

The foundational principle of utility-scale Renewable Energy Funding is the creation of a bankruptcy-remote Special Purpose Vehicle (SPV). Unlike traditional corporate balance sheet loans, where lenders look to the general creditworthiness and unencumbered assets of the corporate parent, SPV project finance isolates the renewable energy project as an independent financial entity. Lenders rely primarily on the project's operational cash flows generated under long-term Power Purchase Agreements (PPAs) to service debt, while securing a first-ranking charge on project land, solar modules, wind turbine generators (WTGs), sub-stations, transmission lines, and escrowed bank accounts.

Under a non-recourse structure, the corporate sponsor's liability is strictly limited to its equity commitment and specific completion guarantees prior to the Date of Commercial Operation (COD). Once COD is achieved and the project demonstrates stabilized generation, sponsor recourse falls away completely, allowing corporate developers to undertake multi-gigawatt development pipelines without over-leveraging their corporate balance sheets.

1.2 The Renewable Energy Capital Mismatch Paradox

Renewable energy projects exhibit a unique cash flow profile known as the *Renewable Energy Capital Mismatch Paradox*. A typical 100 MW solar or wind project requires a massive upfront capital expenditure (CAPEX) of ₹450 Crores to ₹650 Crores during its 12-to-24-month construction phase, during which the project generates zero operating revenue. Upon commissioning, the project enters an operational phase lasting 25 to 30 years with minimal operational expenditure (OPEX)—typically 1.0% to 1.5% of initial CAPEX annually—and zero fuel cost.



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Commercial lenders cannot evaluate renewable energy projects using short-term working capital or traditional 5-year corporate loan metrics. Instead, debt facilities must be structured with construction-phase principal moratoriums, long operational door-to-door tenures (15 to 20 years), and sculpted debt service schedules aligned with seasonal generation profiles (monsoon wind peaks, summer solar peaks) and PPA tariff payment cycles.

1.3 National Net-Zero Targets, Regulatory Framework & RBI Guidelines

India's commitment to achieving 500 GW of non-fossil fuel energy capacity by 2030 and net-zero carbon emissions by 2070 forms the macro-economic backdrop for green credit expansion. Key institutions driving this capital allocation include the Indian Renewable Energy Development Agency (IREDA), Power Finance Corporation (PFC), REC Limited, State Bank of India (SBI), and international multilateral institutions like the International Finance Corporation (IFC) and Asian Development Bank (ADB).

The regulatory framework is governed by: - **Electricity Act, 2003 & CERC/SERC Regulations:** Establishing competitive bidding guidelines, tariff determination mechanisms, and Renewable Purchase Obligations (RPO) enforced on DISCOMs and captive consumers. - **Green Energy Open Access Rules:** Liberalizing open access for C&I consumers above 100 kW, capping cross-subsidy and additional surcharges to accelerate Group Captive and Third-Party PPAs. - **RBI Project Finance Directions (2025/2026):** Mandating a 5.0% standard asset provision during the construction phase of infrastructure projects, stepping down to 2.5% upon COD and 1.0% after two years of operational cash flow stability, while enforcing strict 2-year Date of Commercial Operation (DCCO) extension caps.

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Module 2: Major Financing Modalities & Product Structures

2.1 Utility-Scale Greenfield Project Finance (BOO / BOOT)

Utility-scale projects involve Build-Own-Operate (BOO) or Build-Own-Operate-Transfer (BOOT) structures developed under central or state tenders issued by sovereign intermediaries like the Solar Energy Corporation of India (SECI), NTPC, or NHPC. Projects benefit from 25-year fixed-tariff PPAs backed by tri-partite payment security mechanisms (State Government, RBI, and Union Ministry of Power).

Debt for utility-scale greenfield projects is underwritten at a Debt-to-Equity ratio of 70:30 to 80:20. Lenders disburse debt in LIE-certified (Lender's Independent Engineer) tranches aligned with physical construction milestones (land acquisition, boundary fencing, civil foundations, module/WTG delivery, sub-station erection, and grid synchronization).

2.2 Commercial & Industrial (C&I) Rooftop & Open Access Solutions

Commercial & Industrial (C&I) projects supply clean power directly to corporate manufacturing plants, IT parks, and commercial complexes under two primary structures: 1. **Group Captive Structure:** Under the Electricity Rules 2005, corporate off-takers hold at least 26% equity in the SPV and consume at least 51% of generated power, qualifying for complete exemption from Cross-Subsidy Surcharges (CSS) and Additional Surcharges (AS). 2. **Third-Party Open Access PPA:** The SPV sells power directly to industrial buyers through state transmission grids, subject to prevailing SERC open access surcharges.

C&I projects offer higher operational yields (PPA tariffs of ₹4.50 to ₹6.50/kWh vs. SECI utility tariffs of ₹2.50 to ₹3.00/kWh), but lenders apply shorter debt tenures (10 to 12 years) and mandate strict corporate off-taker credit assessments (CIBIL CMR 1 to 4, investment-grade rating).

2.3 Hybrid & Energy Storage Systems (Solar + Wind + BESS)

To overcome the intermittency of standalone solar (available during daylight) and wind (peaking during specific months and night hours), the market has shifted toward Hybrid Solar-Wind and Battery Energy Storage Systems (BESS) / Pumped Storage Hydro (PSH). These projects provide Round-The-Clock (RTC) or Peak-Power supply.

Financing BESS and Hybrid projects involves underwriting dual generation profiles and battery degradation replacement capex (Augmentation CAPEX in Year 8–10). Lenders structure specialized Battery Replacement Reserve Accounts (BRRA) funded from operational cash flows to ensure battery stack replacement without refinancing default.

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2.4 Green Bonds, NCDs & InvIT Refinancing Take-Out

Once a greenfield renewable project achieves COD and establishes a 12-to-24-month operational track record, corporate developers execute capital recycling via: - **Infrastructure Investment Trusts (InvITs):** Transferring operational yielding assets to public or private InvITs (e.g., IndiGrid, Vena Energy InvIT) to unlock equity capital for new development. - **Rupee Green Bonds & NCDs:** Issuing rated Non-Convertible Debentures (NCDs) or Rupee Denominated Bonds (Masala Bonds) to domestic pension funds and insurance companies at lower interest rates (7.50% to 8.75% p.a.) over 15-to-20-year tenures, paying off high-cost construction debt.

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Module 3: Comparative Capital Architecture Matrix

Parameter / Feature	Utility-Scale SPV Debt (SECI / State)	C&I Group Captive Project Debt	Rooftop Solar Capex Loan	Green Bond / InvIT Refinancing	Private Credit / Mezzanine AIF Debt
Primary Borrower Entity	Standalone Bankruptcy-Remote SPV	C&I SPV (26% Off-taker Equity)	Corporate Operating Company	Listed / Private InvIT or SPV	Developer Parent HoldCo / SPV
Off-Take Agreement	25-Year Sovereign PPA (SECI/NTPC)	10-25 Year Group Captive PPA	Self-Consumption / Net Metering	Portfolio Operating PPAs	Equity-Linked / Subordinated Cash
Target Debt-to-Equity Ratio	75:25 to 80:20	70:30 to 75:25	65:35 to 70:30	75:25 (InvIT Leverage Cap 70%)	85:15 (Total SPV Debt)
Pricing Benchmark & Spread	EBLR / MCLR + 1.25%-2.25% p.a.	EBLR / MCLR + 1.75%-2.75% p.a.	EBLR / MCLR + 2.00%-3.25 % p.a.	Fixed NCD Coupon (7.50%-8.75%)	Fixed IRR (14.00%-18.00% p.a.)
Construction Moratorium	12-24 Months (DCCO)	9-15 Months	3-6 Months	Zero (Fully Operational Assets)	12-36 Months
Door-to-Door Loan Tenure	15 - 20 Years Post-COD	10 - 15 Years	7 - 10 Years	10 - 20 Years (Bullet/NCD)	3 - 5 Years (Subordinated)
Primary Collateral Base	1st Charge Land, Panels, Sub-station	1st Charge Land/PPA Escrow	Charge on Rooftop Equipment	Pledged Units / Portfolio Assets	2nd Charge / Equity Share Pledge

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Parameter / Feature	Utility-Scale SPV Debt (SECI / State)	C&I Group Captive Project Debt	Rooftop Solar Capex Loan	Green Bond / InvIT Refinancing	Private Credit / Mezzanine AIF Debt
Target DSCR Threshold	Average DSCR >= 1.25x - 1.35x	Average DSCR >= 1.30x - 1.40x	DSCR >= 1.35x	Portfolio DSCR >= 1.30x	Combined DSCR >= 1.15x
Escrow TRA Control Level	Mandatory 7-Tier TRA Waterfall	Mandatory TRA Escrow	Escrow / Auto-Debit Mandate	Master InvIT Cash Sweep TRA	Subordinated TRA Cash Sweep
Tax Shield Applicability	Sec 32 (40% WDV Dep) + Sec 36(1)(iii)	Sec 32 + Sec 36(1)(iii)	Sec 32 + Sec 36(1)(iii)	Sec 194LBA Tax Pass-Through	Sec 36(1)(iii) Interest Deduction

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Module 4: Underwriting Framework, Tariff Off-Take Structures & Sizing Formulas

4.1 Quantitative Debt Sizing Metrics & Formulas

Project finance lenders apply quantitative metrics to size the maximum permissible debt limit for a renewable energy asset:

1. Cash Flow Available for Debt Service (CFADS)

CFADS represents the true operational cash generated by the SPV available to meet principal and interest obligations:

$$CFADS_t = EBITDA_t + Interest\ Income_t - Taxes\ Paid_t - Maintenance\ CAPEX_t - \Delta NWC_t$$

Where: -

$EBITDA_t = PPA\ Tariff\ Revenue - O\&M\ Costs - Insurance\ Premiums - Statutory/Grid\ Charges$

- ΔNWC_t = Change in Net Working Capital (accounting for DISCOM payment lags).

2. Debt Service Coverage Ratio (DSCR)

Lenders evaluate Year-by-Year (Annual) DSCR and Average DSCR across the loan tenure:

$$DSCR_t = \frac{CFADS_t}{Principal\ Amortization_t + Interest\ Expense_t}$$

- **Minimum Annual DSCR Covenant:** $\geq 1.15x$ to $1.20x$ (in worst-case generation year).
- **Average DSCR Target:** $\geq 1.25x$ to $1.35x$ across the 15-to-20-year operational tenure.

3. Loan Life Coverage Ratio (LLCR) & Project Life Coverage Ratio (PLCR)

LLCR measures the Net Present Value (NPV) of CFADS over the remaining loan tenure divided by outstanding debt:

$$LLCR_t = \frac{\sum_{k=t}^{T_{loan}} \frac{CFADS_k}{(1+i)^k}}{Outstanding\ Debt_t} \geq 1.35x$$

PLCR measures NPV over the full economic life of the asset (25 years):

$$PLCR_t = \frac{\sum_{k=t}^{T_{asset}} \frac{CFADS_k}{(1+i)^k}}{Outstanding\ Debt_t} \geq 1.50x$$

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4.2 Generation Yield Modeling: P50, P90, and P99 Estimates

Solar and wind energy generation is naturally variable due to solar irradiance (GHI/GTI) and wind velocity fluctuations. Lenders commission independent Solar / Wind Resource Assessment (SRA/WRA) reports from certified agencies (e.g., DNV, TÜV Rheinland, Mott MacDonald, Gensol) to establish probability-based generation estimates:

- **P50 Generation Level:** The generation estimate exceeded with a 50% probability (median expectation). Equity promoters use P50 to calculate expected equity Internal Rate of Return (IRR).
- **P90 Generation Level:** The generation estimate exceeded with a 90% probability (conservative downside case). **Lenders strictly size debt capacity and DSCR covenants against P90 generation figures.**
- **P99 Generation Level:** The extreme downside scenario (99% confidence level) used for stress-testing minimum debt service survival without default.

RESOURCE ASSESSMENT GENERATION PROBABILITY CURVE		
Probability Level	**Description**	**Application in Renewable Finance**
P50 (50% Prob)	Median Generation	Promoter Equity Returns (Base Model)
P90 (90% Prob)	Conservative Yield	BANK DEBT SIZING & DSCR COVENANTS
P99 (99% Prob)	Stress Downside Yield	Liquidity Buffer & DSRA Sizing Audit

4.3 Capacity Utilization Factor (CUF / PLF) & Degradation Math

Generation modeling incorporates the annual Capacity Utilization Factor (CUF):

$$Annual\ Energy\ Generation\ (kWh) = Installed\ Capacity\ (kW) \times 8,760\ Hours \times CUF\ (\%)$$

For Solar PV: - Typical Solar CUF in India: 21% to 26% (depending on location: Rajasthan/Gujarat ~24-26%; South India ~21-23%). - **Panel Degradation Factor:** Linear degradation modeled at 2.0% in Year 1, and 0.55% per annum from Year 2 to Year 25 (guaranteeing 80%+ output at Year 25).

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For Wind Energy: - Typical Wind CUF (WTG hub heights 120m–140m): 30% to 38%. - **Wind Degradation & Availability:** WTG operational availability modeled at 95% to 97%, with 0.5% annual mechanical degradation.

4.4 Off-Take Tariff Security & Payment Security Mechanisms (PSM)

The credit rating of the power purchaser (off-taker) dictates credit pricing and collateral requirements: 1. **SECI / NTPC / NHPC (AAA Sovereign Counterparties):** Power purchase agreements include a Payment Security Mechanism (PSM) comprising a 1-month revolving Letter of Credit (LC), a Payment Security Fund maintained by SECI, and state government guarantees. 2. **State DISCOMs (e.g., TANGEDCO, Rajasthan DISCOMs, UPPCL):** Due to historic DISCOM payment delays (6 to 12 months), lenders mandate: - A 3-to-6-month Debt Service Reserve Account (DSRA). - Higher pricing spreads (+100 to +200 bps). - Enforcing Late Payment Surcharge (LPS) Rules under the Electricity Act. 3. **C&I Corporate Off-Takers:** Off-taker must hold an investment-grade rating (CRISIL/ICRA BBB or better). The PPA mandates a Termination Compensation Clause where the off-taker must buy out the plant or pay full debt service if the agreement is terminated prematurely.

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Module 5: Security Architecture, Escrow TRA Waterfalls & Statutory Covenants

5.1 Primary Collateral & Contractual Assignment Architecture

Renewable energy debt facilities require a comprehensive, multi-layered security package covering physical, financial, and contractual assets:

1. **Equitable Mortgage over Project Land & Rights (Section 58(f) Transfer of Property Act 1882):** First exclusive charge on freehold land or long-term leasehold rights (25–30 year lease agreements with State Industrial Development Corporations or private landowners, with registered lease deeds and sub-leasing NOCs).
2. **First Exclusive Charge on Fixed Assets & Movables:** Hypothecation of all solar modules, mounting structures (trackers/fixed-tilt), wind turbine generators (WTGs), transformers, inverters, BESS battery racks, transmission towers, bay extensions, and sub-station equipment under Section 2(1)(n) of the SARFAESI Act 2002.
3. **Absolute Contractual Assignment of Project Agreements:** Irrevocable assignment of:
   - Power Purchase Agreement (PPA) and Off-take Letters of Credit.
   - Engineering, Procurement, and Construction (EPC) Lump-Sum Turnkey Contracts (including Liquidated Damages LD provisions for performance/delay).
   - Long-Term Operation & Maintenance (O&M) Contracts.
   - Transmission Interconnection Agreements with STU/CTU (Power Grid Corporation of India - PGCIL).
   - Inverter, Transformer, and WTG OEM Performance Warranties (e.g., 25-year solar module performance warranty; 10-year WTG warranty).
   - Comprehensive All-Risk Marine, Erection, Operational Transit, and Business Interruption Insurance Policies.
4. **Pledge of Equity Shares:** Pledge of 51% to 100% of the voting equity shares of the SPV held by the corporate sponsor, alongside irrevocable Powers of Attorney (POA) enabling lenders to enforce share pledge and execute sponsor substitution upon default under Section 172/176 of the Indian Contract Act 1872.

5.2 Mandatory Statutory Filings: CERSAI, ROC Form CHG-1 & NeSL

Under Section 77 of the Companies Act 2013, the security interest over SPV assets must be registered with the Registrar of Companies (ROC) by filing **Form CHG-1** within 30 days of security creation. Furthermore, charges are registered on the **CERSAI Portal** (Central Registry of Securitisation Asset Reconstruction and Security Interest) to establish legal priority over

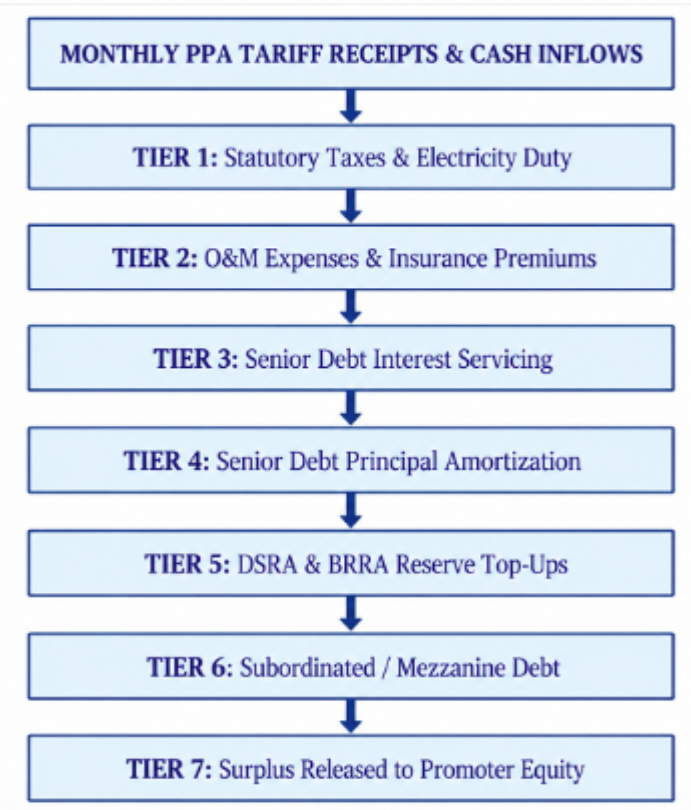
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third-party creditors, and filed with **NeSL** (National E-Governance Services Limited) as an Information Utility under the Insolvency and Bankruptcy Code (IBC) 2016.

MANDATORY STATUTORY CHARGE CREATION TIMELINE	
Day 0:	Execution of Loan Agreement, Hypothecation Deed & Mortgage Security
Day 1-30:	Filing ROC Form CHG-1 under Section 77 Companies Act 2013
Day 1-30:	CERSAI Portal Charge Registration for First-Charge Asset Priority
Day 1-30:	NeSL Information Utility Authentication under IBC 2016

5.3 The 7-Tier Trust and Retention Account (TRA) Escrow Waterfall

To ensure complete cash flow ring-fencing, 100% of PPA billing collections, insurance proceeds, and tariff payments are deposited directly into a bank-controlled Trust and Retention Account (TRA) held with the Lead Monitoring Bank. Cash distributions follow a strict statutory 7-tier waterfall:



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1. **Tier 1 (Statutory Outflows):** GST, Electricity Duty, land lease rents, and statutory taxes.
2. **Tier 2 (Essential OPEX):** LIE-approved monthly Operation & Maintenance (O&M) expenses, grid transmission charges, and insurance premiums.
3. **Tier 3 (Senior Debt Interest):** Monthly interest payment on senior project loans.
4. **Tier 4 (Senior Debt Principal):** Scheduled principal amortization installments.
5. **Tier 5 (Reserve Top-Ups):** Funding the Debt Service Reserve Account (DSRA - 1 to 2 quarters P+I) and Battery Replacement Reserve Account (BRRA for storage projects).
6. **Tier 6 (Subordinated / Junior Debt):** Interest and principal on subordinated mezzanine facilities.
7. **Tier 7 (Promoter Surplus Sweep):** Residual free cash flow released to corporate sponsor equity **only if DSCR covenants (e.g., >= 1.20x) and DSRA requirements are fully satisfied**.

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Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

6.1 Indicative Tenures & Construction Moratoriums

Renewable energy term loans offer extended tenures matching the long-term nature of PPAs: - **Construction Phase (DCCO):** 12 to 24 months, during which interest is capitalized into Capital Work-in-Progress (CWIP) under Ind AS 23 or paid monthly. - **Operational Amortization Phase:** 15 to 20 years post-COD, structured as quarterly or semi-annual amortizing installments. - **Tail Period:** Lenders mandate a minimum 3-to-5-year "tail period" between loan maturity (e.g., Year 18–20) and PPA expiration (Year 25), providing a safety cushion for debt refinancing or extension if generation underperforms.

6.2 Commercial Pricing Mechanics & Floating Rate Benchmarking

Loans are priced against floating benchmarks (EBLR / MCLR / 3-Month T-Bill) plus a risk-adjusted credit spread:

$$All - In\,Borrowing\,Cost = Floating\,Benchmark\,(e.g.,\ 1 - Yr\,MCLR\,@\,8.25\%) + Spread\,(1.00\%to$$

- **SECI/NTPC Utility Projects:** Spread ranges from 1.00% to 1.50% p.a., resulting in an all-in rate of 9.25% to 9.75% p.a.
- **C&I Group Captive Projects:** Spread ranges from 1.75% to 2.50% p.a., resulting in an all-in rate of 10.00% to 10.75% p.a.
- **Reset Clauses:** Spreads are reset every 1 to 3 years based on project operational performance and credit rating upgrades.

6.3 Section 32 Accelerated Depreciation & Section 36(1)(iii) Tax Shields

Renewable energy investments benefit from substantial tax incentives under the Income Tax Act, 1961:

1. **Section 32 Accelerated Depreciation:** Solar panels, wind turbines, and pollution control equipment qualify for a **40% Written Down Value (WDV) depreciation rate** in Year 1 (or 20% if commissioned in the second half of the fiscal year). For corporate developers with profitable parent operations, this creates a massive tax shield against taxable profits.
2. **Section 36(1)(iii) Debt Interest Deduction:** Interest paid on project debt is 100% tax-deductible as a business expense. At the corporate tax rate of 25.168% (inclusive of surcharge and cess under Section 115BAA), the effective post-tax cost of debt is significantly lower than the nominal interest rate:

$$Effective\,Post - Tax\,Interest\,Rate = Nominal\,Rate \times \left(1 - T_{corp}\right) = 9.50\% \times (1 - 0.25168) = 7.11\%\,p.a.$$

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SECTION 32 ACCELERATED DEPRECIATION TAX SHIELD IMPACT	
Asset Value:	₹100 Crores Solar Equipment
Year 1 Tax Depreciation (40% WDV):	₹40 Crores
Tax Shield Cash Savings @ 25.168% Corporate Tax Rate:	₹10.07 Crores
Impact:	Direct upfront cash liquidity boost for promoter capital recycling

6.4 Ind AS 16 & Ind AS 23 Accounting Standards

- **Ind AS 16 (Property, Plant & Equipment):** Mandatory component accounting, where major components (inverters, WTG gearboxes, solar panels) are depreciated separately over their useful lives.
- **Ind AS 23 (Borrowing Costs):** Borrowing costs directly attributable to the acquisition and construction of a "qualifying asset" are capitalized into the carrying cost of CWIP until the asset achieves COD. Upon COD, capitalization ceases and interest is expensed directly to the Profit & Loss statement.

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Module 7: Suitable Sectors, Borrower Profiles & Market Alignment

7.1 Target Developer & Borrower Profiles

Renewable Energy Funding is structured for four primary market segments: 1. **Utility-Scale Independent Power Producers (IPPs):** Institutional developers (e.g., Tata Power Renewables, ReNew Power, Adani Green, ACME Solar, Hero Future Energies) executing multi-hundred MW solar, wind, and hybrid projects under SECI/NTPC central tenders. 2. **C&I Open Access Developers:** Specialized developers focused on rooftop and ground-mounted Group Captive projects for industrial clients (auto OEMs, pharma, steel, data centers, chemical plants). 3. **Turnkey EPC Contractors:** Engineering firms requiring non-fund-based Bank Guarantees (Performance BGs, Advance BGs) and Inland Usance LCs to execute EPC packages for IPPs and public utilities. 4. **Captive Industrial Consumers:** Manufacturing conglomerates investing in rooftop solar or captive wind capacity to lower electricity tariffs and meet ESG Net-Zero targets.

7.2 Market Footprint & Institutional Financing Ecosystem

According to Indian banking data, renewable energy credit exposure across Indian Commercial Banks, Public Financial Institutions (IREDA, PFC, REC), and NBFCs exceeds **₹4.5 Lakh Crores**, representing one of the fastest-growing credit segments. Digital supply chain platforms such as SBI's **e-DFS (₹49,261 Crores)** and **e-VFS (₹28,092 Crores)** support vendor financing for solar panel and inverter distributors, while IREDA provides specialized debt facilities for green hydrogen and BESS projects.

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Module 8: Key Advantages and Strategic Limitations

8.1 Key Strategic Advantages

1. **predictable Cash Flow Security:** Fixed 25-year PPA tariffs eliminate price volatility, guaranteeing stable revenues protected against market inflation.
2. **Bankruptcy-Remote Off-Balance-Sheet Sizing:** Non-recourse SPV structures preserve corporate parent borrowing limits for parallel expansion.
3. **Substantial Tax Shield Benefits:** Section 32 (40% WDV accelerated depreciation) and Section 36(1)(iii) interest deductions significantly lower post-tax capital cost.
4. **Long-Term Amortization Matching Asset Life:** 15-to-20-year operational loan tenures prevent refinancing cliffs and support smooth equity returns.
5. **ESG Capital Access:** Access to global green bond markets, ESG funds, and multilateral debt at competitive interest rates.

8.2 Strategic Limitations & Risk Mitigation

1. **Weather Intermittency & Resource Risk:** Seasonal generation fluctuations (monsoons, low-wind periods). *Mitigation: Underwrite debt against P90 generation estimates and mandate a 2-quarter DSRA.*
2. **State DISCOM Payment Delays & Contract Renegotiation Risk:** Historical delays by state utilities. *Mitigation: Focus on SECI/NTPC PPAs or enforce Late Payment Surcharge (LPS) rules and C&I Group Captive structures.*
3. **Grid Curtailment & Transmission Bottlenecks:** Lack of sub-station evacuation capacity. *Mitigation: Require LIE certification of Stage-2 PGCIL/STU grid connectivity prior to financial closure.*
4. **Foreign Exchange Risk on Imported Equipment:** Solar panel and BESS imports exposed to USD/INR fluctuations. *Mitigation: Mandate 100% FX hedging via Forward Covers or Cross-Currency Swaps (CCS) for import LCs/ECBs.*

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Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)

9.1 Project Profile: “Mumbai Green-Power Infra SPV Ltd. - MGPIPL”

To demonstrate the quantitative mechanics of Renewable Energy Funding, consider a 250 MW Greenfield Hybrid Solar-Wind Project in Jaisalmer, Rajasthan developed by *Mumbai Green-Power Infra SPV Ltd. (MGPIPL)*, a bankruptcy-remote SPV owned by an institutional power developer.

Key Project Parameters:

- **Capacity Composition:** 150 MW Solar PV + 100 MW Wind Energy (Total 250 MW Hybrid).
- **Off-Take PPA:** 25-Year Fixed Tariff PPA with SECI @ **₹2.80 / kWh**.
- **Total Project Outlay (TPO): ₹1,250.00 Crores** (Solar CAPEX ₹600 Cr + Wind CAPEX ₹500 Cr + Sub-station/Grid ₹100 Cr + IDC/Pre-op ₹50 Cr).
- **Construction Horizon (DCCO):** 18 Months (capitalize interest under Ind AS 23).
- **Annual Energy Generation (P90 Estimate):** 613.20 Million Units (kWh) / year (Combined Hybrid CUF = 28.0%).
- **Annual Gross PPA Revenue:** 613.20 Million kWh × ₹2.80/kWh = **₹171.70 Crores / year**.
- **Annual Operating Expenses (O&M, Insurance, Grid Fees): ₹21.70 Crores / year**.
- **Annual Net Operating EBITDA:** ₹171.70 Cr - ₹21.70 Cr = **₹150.00 Crores / year**.

9.2 Structural Comparison: Pathway A vs. Pathway B

The developer evaluates two financing options: - **Pathway A (Corporate Balance Sheet Term Loan):** 60:40 Leverage (Debt: ₹750 Cr @ 12.00% p.a.; Corporate Equity: ₹500 Cr). 5-year straight amortization (₹150 Cr/year principal) with zero construction moratorium. - **Pathway B (Structured Non-Recourse SPV Renewable Energy Project Debt):** 75:25 Leverage (Debt: ₹937.50 Cr @ 9.25% p.a.; SPV Equity: ₹312.50 Cr). 18-month construction principal moratorium + 18-year operational sculpted amortization (₹52.08 Cr/year avg. principal) + Section 32 tax shields.

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FINANCING PATHWAY COMPARISON SUMMARY		
Parameter	**Pathway A (Corporate Loan)**	**Pathway B (SPV Debt)**
SPV Debt Sized	₹750.00 Crores	₹937.50 Crores
Promoter Cash Equity Required	₹500.00 Crores	₹312.50 Crores
Upfront Cash Equity Preserved	BASELINE	+₹187.50 CRORES
Interest Rate (p.a.)	12.00% p.a.	9.25% p.a.
Principal Moratorium	ZERO	18 MONTHS (DCCO)
Operational Repayment Tenure	5 Years	18 Years
Year 1 Operational DSCR	0.625x (SEVERE DEFAULT)	1.398x (INVESTMENT)

9.3 Detailed Financial Model & Debt Servicing Schedule (INR Crores)

Pathway A (Corporate Balance Sheet Loan):

- Year 1 Construction Cash Debt Service Outflow: Interest ₹90.00 Cr + Principal ₹150.00 Cr = **₹240.00 Crores**.
- Year 1 Operational EBITDA: **INR 0.00** (under construction). Result: **₹240.00 Crore Year-1 Cash Deficit (CORPORATE DEFAULT)**.
- Year 2 (Post-COD Operational Phase):
  - Net EBITDA: **₹150.00 Crores**.
  - Debt Service: Interest ₹90.00 Cr + Principal ₹150.00 Cr = **₹240.00 Crores**.
  - Operational DSCR: $ ext{DSCR} = rac{ ext{₹150.00 Cr}}{ ext{₹240.00 Cr}} = $ (**FAILING INSOLVENCY DEFAULT**).

Pathway B (Structured Non-Recourse Renewable Energy SPV Debt):

- Construction Phase (Months 1–18): Interest capitalized into CWIP under Ind AS 23. **Year 1 Cash Outflow = INR 0.00 (100% SOLVENT)**.
- Year 2 (Post-COD Operational Year 1):
  - Net EBITDA: **₹150.00 Crores**.
  - Senior Debt Principal Installment: **₹21.00 Crores** (sculpted step-up amortization).
  - Senior Debt Interest Expense (@ 9.25% p.a. on ₹937.50 Cr): **₹86.72 Crores**.
  - Total Senior Debt Service: ₹21.00 Cr + ₹86.72 Cr = **₹107.72 Crores**.
  - Operational DSCR: $ ext{DSCR} = rac{ ext{₹150.00 Cr}}{ ext{₹107.72 Cr}} = $ (**PRIME INVESTMENT GRADE COVER**).

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PATHWAY B 5-YEAR CASH FLOW & DSCR WATERFALL (INR CRORES)						
Year	**EBITDA**	**Interest**	**Principal**	**Total Debt Service**	**CFADS**	**DSCR**
Year 1	0.00*	0.00*	0.00	0.00	0.00	N/A
Year 2	150.00	86.72	21.00	107.72	150.00	1.392x
Year 3	149.18**	84.78	26.25	111.03	149.18	1.344x
Year 4	148.36	82.35	31.50	113.85	148.36	1.303x
Year 5	147.54	79.43	36.75	116.18	147.54	1.270x

* Interest capitalized under Ind AS 23 during 18-month construction period.

** EBITDA accounts for 0.55% annual solar/wind panel degradation.

9.4 Opportunity Cost, Tax Shield & Net Wealth Surplus Analysis

1. **Promoter Cash Equity Preserved:** Pathway B requires ₹312.50 Crores equity vs ₹500.00 Crores in Pathway A, unlocking **₹187.50 Crores in liquid equity**.
2. **Reinvestment Income on Preserved Equity:** Reinvesting ₹187.50 Crores into parallel renewable projects earning 16.00% p.a. ROCE over 5 years generates a net post-tax return of **+₹112.50 Crores**.
3. **Section 32 Accelerated Depreciation Tax Shield:** In Year 1 post-COD, MGPIPL claims 40% WDV depreciation on ₹1,100 Crores equipment value = ₹440 Crores depreciation. At a 25.168% tax rate, this yields an upfront tax cash saving of **+₹110.74 Crores**.
4. **Section 36(1)(iii) Debt Interest Tax Shield:** Average annual interest of ₹83.32 Crores yields an annual tax shield of ₹20.97 Crores (Total 5-year tax shield = **+₹104.85 Crores**).
5. **Effective Post-Tax Interest Rate:** Slashed from nominal 9.25% p.a. down to **6.92% p.a.**
6. **Total Opportunity-Adjusted Net Wealth Surplus (5-Year Horizon):**

$$Net\ Wealth\ Surplus = Equity\ Preserved\ (₹187.50\ Cr) + ROCE\ Reinvestment\ (₹112.50\ Cr) + Tax\ Shields\ (₹215.59\ Cr) - Bank\ Setup\ Fees\ (₹9.38\ Cr) =+ ₹506.21\ CRORES$$

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5-YEAR STRATEGIC WEALTH SURPLUS BREAKDOWN	
Upfront Cash Equity Preserved	+₹187.50 Crores
Reinvestment Income on Preserved Equity (@ 16%)	+₹112.50 Crores
Section 32 & 36(1)(iii) Combined Tax Shields	+₹215.59 Crores
Less Upfront Processing & Structuring Fees (1%)	-₹9.38 Crores
TOTAL NET FINANCIAL WEALTH SURPLUS GENERATED	**+₹506.21 CRORES**

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Module 10: Regulatory Compliance, RBI IRACP Norms & Default Framework

10.1 RBI Project Finance Guidelines (2025/2026 Directives)

Under the updated Reserve Bank of India (Project Finance) Directions, infrastructure and renewable energy lending institutions must adhere to strict provisioning and monitoring guidelines: 1. **Construction Phase Provisioning:** Standard asset provisioning for greenfield project loans during the construction phase is set at **5.0%** (up from 0.40%), reflecting construction phase execution risks. 2. **Post-COD Provision Step-Down:** Upon achieving commercial operation (COD), provisioning steps down to **2.5%**. Once the project demonstrates 24 months of operational stability with a minimum DSCR of $1.25x$, provisioning reverts to the standard **1.0%**. 3. **DCCO Extension Limits:** The Date of Commercial Operation (DCCO) can be extended by up to **2 years** for court litigation, land acquisition delays, or grid interconnection delays without mandatory NPA classification, provided debt restructuring terms are executed.

10.2 Income Recognition, Asset Classification & Restructuring (IRACP Norms)

In accordance with RBI IRACP norms, a renewable energy loan is classified as a Non-Performing Asset (NPA) if interest or principal installment remains overdue for **more than 90 days**: - **SMA-0:** Overdue 1 to 30 days. - **SMA-1:** Overdue 31 to 60 days. - **SMA-2:** Overdue 61 to 90 days. - **NPA (Sub-Standard):** Overdue > 90 days.

RBI IRACP STRESS & NPA CLASSIFICATION TIMELINE	
Day 1 - 30:	SMA-0 Stress Trigger (Early Alert & Monitoring Bank Notification)
Day 31 - 60:	SMA-1 Trigger (Mandatory Escrow TRA Cash Flow Review)
Day 61 - 90:	SMA-2 Trigger (Formulation of Resolution Plan / Intercreditor Agreement)
Day 91+:	Non-Performing Asset (NPA) Classification & SARFAESI Sec 13 Enforcement

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10.3 Legal Default Remedies & Enforcement Mechanisms

If an SPV defaults on debt service, lenders exercise legal remedies under Indian law: 1. **SARFAESI Act 2002 (Section 13/14):** Issuing a 60-day demand notice under Section 13(2), taking physical possession of SPV assets, solar panels, and sub-stations under Section 13(4) with District Magistrate assistance under Section 14. 2. **Sponsor Substitution Rights:** Invoking PPA and TRA step-in rights to replace the defaulting developer with a qualified third-party renewable IPP without terminating the 25-year SECI/state PPA. 3. **Insolvency and Bankruptcy Code (IBC) 2016:** Initiating Corporate Insolvency Resolution Process (CIRP) before the National Company Law Tribunal (NCLT) under Section 7. 4. **Section 138 NI Act & Section 25 PSSA:** Initiating criminal proceedings for dishonored post-dated cheques or auto-debit escrow mandates.

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Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test absolute mastery over Renewable Energy Funding.

Question 1

A 100 MW solar SPV has an annual EBITDA of ₹60.00 Crores, annual debt interest of ₹30.00 Crores, and scheduled principal amortization of ₹15.00 Crores. What is the SPV's Debt Service Coverage Ratio (DSCR)? A) 1.20x
B) 1.33x
C) 1.50x
D) 2.00x

Question 2

Why do project finance lenders size renewable energy debt facilities against P90 generation estimates rather than P50 estimates? A) P50 represents an unrealistically high downside stress scenario.
B) P90 provides a conservative 90% probability downside yield, protecting debt service against weather intermittency.
C) P90 allows promoters to maximize equity dividends in Year 1.
D) P50 is legally prohibited under CERC regulations.

Question 3

Under a non-recourse Special Purpose Vehicle (SPV) project finance structure, what is the primary source of debt repayment? A) Unencumbered balance sheet assets of the corporate parent promoter.
B) Unsecured personal guarantees issued by corporate directors.
C) Project operational cash flows ring-fenced in a Trust and Retention Account (TRA) Escrow under a PPA.
D) Annual equity infusions from short-term venture capital funds.

Question 4

Under Section 32 of the Income Tax Act 1961, what is the allowable Written Down Value (WDV) accelerated depreciation rate for solar PV modules and wind turbines? A) 15%
B) 25%

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C) 40%
D) 80%

Question 5

In the 7-tier Trust and Retention Account (TRA) escrow waterfall, which outflow takes priority over Senior Debt Interest Servicing? A) Subordinated Mezzanine Debt Interest.
B) Promoter Dividend Distributions.
C) Statutory Taxes, Land Lease Rents, O&M Expenses, and Insurance Premiums.
D) Battery Replacement Reserves.

Question 6

Under the Electricity Rules 2005, what are the two mandatory criteria required for a project to qualify as a Group Captive structure? A) Off-taker holds 51% equity and consumes 100% of power.
B) Off-taker holds at least 26% equity in the SPV and consumes at least 51% of generated power.
C) Off-taker holds 10% equity and consumes 25% of power.
D) Off-taker holds 100% equity and sells 50% to SECI.

Question 7

What is the primary function of a Debt Service Reserve Account (DSRA) in a renewable energy project loan? A) Funding the upfront land acquisition cost prior to financial closure.
B) Setting aside 1 to 2 quarters of principal and interest payments in a liquid escrow account to cushion temporary generation shortfalls.
C) Paying annual performance bonuses to EPC contractors.
D) Refinancing foreign currency ECBs upon maturity.

Question 8

According to Section 77 of the Companies Act 2013, within how many days must a company register a charge created on SPV assets (Form CHG-1) with the Registrar of Companies? A) 15 Days
B) 30 Days
C) 60 Days
D) 90 Days

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Question 9

If a company's nominal borrowing cost is 9.50% p.a. and the corporate tax rate is 25.168%, what is the effective post-tax cost of debt after Section 36(1)(iii) interest tax shields? A) 6.50% p.a.
B) 7.11% p.a.
C) 8.25% p.a.
D) 9.50% p.a.

Question 10

Under updated RBI Project Finance Directives (2025/2026), what is the standard asset provisioning requirement for greenfield infrastructure project loans during the active construction phase? A) 0.40%
B) 1.00%
C) 2.50%
D) 5.00%

Answer Key & Detailed Mathematical / Legal Rationales

Answer 1: B) 1.33x

Rationale:

$$DSCR = \frac{EBITDA}{Principal\ Amortization + Interest\ Expense} = \frac{₹60.00\ Cr}{₹15.00\ Cr + ₹30.00\ Cr} = \frac{₹60.00\ Cr}{₹45.00\ Cr} = 1.333x$$

A DSCR of 1.33x satisfies standard bank underwriting covenants ($\geq 1.25x$).

Answer 2: B) P90 provides a conservative 90% probability downside yield, protecting debt service against weather intermittency.

Rationale:
Solar and wind energy generation fluctuates due to weather conditions. P90 represents the generation level exceeded with 90% statistical probability (a conservative yield figure). Sizing debt against P90 ensures the SPV maintains full debt service coverage even during low-solar or low-wind years.

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Answer 3: C) Project operational cash flows ring-fenced in a Trust and Retention Account (TRA) Escrow under a PPA.

Rationale:
In non-recourse project finance, the SPV is bankruptcy-remote from the parent promoter. Lenders do not rely on parent balance sheet assets, but rather on ring-fenced monthly tariff collections paid under long-term PPAs into a bank-controlled TRA Escrow Account.

Answer 4: C) 40%

Rationale:
Under Section 32 of the Income Tax Act 1961, renewable energy generation equipment (solar modules, wind turbine generators, sub-station electronics) qualifies for a 40% Written Down Value (WDV) accelerated depreciation rate, enabling significant tax deferral.

Answer 5: C) Statutory Taxes, Land Lease Rents, O&M Expenses, and Insurance Premiums.

Rationale:
Under the statutory 7-tier TRA Escrow Waterfall, Tier 1 (Statutory Taxes) and Tier 2 (Essential O&M Expenses and Insurance Premiums) must be satisfied before cash flows can be applied to Tier 3 (Senior Debt Interest Servicing) to ensure the physical plant remains operational and compliant.

Answer 6: B) Off-taker holds at least 26% equity in the SPV and consumes at least 51% of generated power.

Rationale:
Rule 3 of the Electricity Rules 2005 stipulates that a power plant qualifies as a Group Captive plant if captive off-taker(s) hold at least 26% of equity ownership and consume at least 51% of the annual electricity generated, exempting them from cross-subsidy surcharges.

Answer 7: B) Setting aside 1 to 2 quarters of principal and interest payments in a liquid escrow account to cushion temporary generation shortfalls.

Rationale:
A Debt Service Reserve Account (DSRA) is a funded cash/FD reserve or SBLC equivalent to 3 to 6 months of debt service obligations, held in escrow to cover short-term liquidity gaps caused by seasonal generation dips or DISCOM billing delays.

Answer 8: B) 30 Days

Rationale:
Section 77(1) of the Companies Act 2013 mandates that every company creating a charge on its assets must register the details with the Registrar of Companies (ROC) in Form CHG-1 within 30 days of creation.

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Answer 9: B) 7.11% p.a.

Rationale:

$$Effective\ Post\ -\ Tax\ Rate\ =\ Nominal\ Rate\ \times \left(1\ -\ T_{corp}\right) = 9.50\% \times (1\ -\ 0.25168) = 7.10904\% \quad (\approx 7.11\%$$

Under Section 36(1)(iii) of the Income Tax Act 1961, business interest payments are tax-deductible, reducing effective borrowing costs.

Answer 10: D) 5.00%

Rationale:
Under the Reserve Bank of India (Project Finance) Directions 2025/2026, standard asset provisioning for greenfield project loans during active construction is set at 5.00%, stepping down to 2.50% at COD and 1.00% upon demonstrating two years of operational stability with DSCR $\geq\ 1.25x$.

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